Lab 9: Neural networks and foundation models

MATH840 | Time Series | Week 10 — graded lab

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The last round

Your series from Week 4, for the fifth and last time. Three times you built a model for it and three times the hidden horizon told you how well it did: ETS in Lab 5, seasonal ARIMA in Lab 7, a combination in Lab 8. Tonight the two newest families from the lectures challenge **your best** — and you decide, with your own numbers, whether anything new was worth it.

Open the notebook

The template: Open in Colab

Click **Copy to Drive** before you start working, otherwise nothing you write is saved. A CPU runtime is enough; the first run downloads about 1.3 GB of model weights.

The practice session — case 5 is this lab on a public series: Open in Colab · Ukrainian version with results and conclusions

The task

1. **Rebuild your champion** — your Lab 8 submission — on the four rolling windows of Lab 8, and write down your hidden-horizon results from Labs 5, 7 and 8.
2. **Neural networks** from the *Neural Networks for Forecasting* lecture — MLP, N-BEATS, LSTM — trained on your series, on the same windows.
3. **Foundation models** from the *Foundation Models for Time Series* lecture — Chronos-2, TimesFM 2.5, TiRex — zero-shot, on the same windows.
4. **One scoreboard** for all of them, plus one mix of old and new decided before you read it.
5. **Submit** the forecast the scoreboard supports, with an interval checked on unseen windows, and **conclude** against everything you have built for this series.

Not this time: plots of the series, decomposition, ACF. Every plot tonight is about the models.

This one counts

8 points. **Deadline: 23:59 today**, on Moodle. The accuracy score is published the next morning.

What you are marked on

Section	Criteria	Points

Section	Criteria	Points
1	Your champion, rebuilt, and your past results	1
2	Forecast accuracy against your benchmark	3
3	Justification block	3
4	Code quality and reproducibility	1
	Total	**8**

The notebook has one section per row. **Keep those headings.**

🏆 Section 1 — Your champion (1 point)

- `MEMBERS`, `CHAMPION` and `CHAMPION_INTERVAL` hold your **submitted** Lab 8 forecast: its members (your Lab 5 ETS, your Lab 7 ARIMA, MSTL), the ones you averaged, and the interval recipe.
- `PAST` holds the hidden-horizon MASE and skill you were sent after Labs 5, 7 and 8.
- The champion's validation-window MASE **must match your Lab 8 table**. If it does not, say why.
- Did not submit Lab 8? Your champion is the better of your Lab 5 and Lab 7 models. Say so; it costs nothing.

🎯 Section 2 — Forecast accuracy (3 points)

What you may submit: **any one forecaster from your scoreboard, or an equal-weight average of two or three of them**, with an 80% interval — the forecaster's own, or conformal from its errors on the four windows. Chosen on the rolling windows; no estimated weights; no models outside the scoreboard.

Scored from your submission file on the `h` observations after your history that nobody has seen:

$$\text{skill} = \frac{\text{MASE of your forecast}}{\text{MASE of your benchmark}}$$

Skill	Points
below 1 — you beat your benchmark	3
1 to 1.1 — within 10%	2
above 1.1	1, and 3.3 must say why

> ⚠ **Train on your own series only**
>
> The Track B pool cuts several series from the same source, disguised. **Another student's history can contain your**

hidden horizon. A network trained across the other series would be trained on your answer. Not allowed, and not needed: the foundation models bring the "many series" experience with them.

ⓘ **What the pool says**

Measured on all 140 issued series, each model fitted on the full served history and scored on the hidden horizon. ETS and ARIMA are the automatic ones; the networks use the template's settings.

Forecast	Beats the benchmark	Within 10%	Worse	Median skill	Worst skill
`AutoETS` + `AutoARIMA` + Chronos-2	**99%**	1%	0%	**0.785**	1.08
`AutoETS` + `AutoARIMA` (≈ a Lab 8 combination)	100%	—	—	0.805	0.93
Chronos-2	84%	9%	7%	0.819	2.05
TimesFM 2.5	76%	10%	14%	0.837	1.61
TiRex	77%	11%	11%	0.869	2.37
N-BEATS, trained on the one series	57%	12%	31%	0.939	3.02
MLP, trained on the one series	50%	14%	36%	0.997	3.48
LSTM, trained on the one series	46%	9%	45%	1.041	4.14

- **The pool was screened so that `AutoETS` and `AutoARIMA` each beat the benchmark** — so their rows are flattered by construction. The new families were not screened: their rows are what you would expect on fresh data.
- **A network trained on one series is close to a coin flip** against the benchmark: a few hundred observations are little to learn from.
- **A foundation model alone** beats the benchmark on 76–84% of the series, and can lose badly on the rest. **Added to the old models** it gave the best median of all — the Week 9 lesson again.
- Your champion is your own Lab 8 forecast, not the automatic one. The table is a guide, not a promise.

✍ Section 3 — Justification block (3 points)

The part a human reads closely. Roughly a paragraph each, with numbers from your own cells.

Item	Points	What earns them
3.1	1	**Neural networks on your series.** MLP, N-BEATS and LSTM against your champion and the benchmark, window by window. Why they did what they did: how many observations they learned from, what `input_size` means for your series, what a second seed changed. Their cost in seconds. Anything you changed and why.
3.2	1	**Foundation models.** The same comparison for Chronos-2, TimesFM 2.5 and TiRex, and their 80% intervals against the conformal one on unseen windows. Your series is disguised public data: can you rule out that a model saw its source? What would you need to know, and how does that change your trust in the numbers?
3.3	1	**What you submit, against the whole course.** Your Lab 5, 7 and 8 hidden-horizon skills next to tonight's rolling evidence. What you submitted and why; the skill you expect. Which approach you would use if you forecast this series at work every month, and what you would check.

Important

Marked on the reasoning. "Nothing new beat my champion, so I submitted it again" earns full marks if the numbers show it and you explain why. So does "a foundation model won three windows of four, here is why I trust it — or don't".

What the model cards say they were trained on

You do not know your series' source, but you know the families: Track B is cut from public Australian and US economic statistics. What each model publishes about its pretraining data:

Model	Published training data
Chronos-2	Technical report, Table 6: 22 real datasets (M4 monthly, electricity and others) plus synthetic data
TimesFM 2.5	Model card: GIFT-Eval pretraining set (GIFT-Eval paper, Table 14: Monash datasets incl. M1, M3, tourism, Australian electricity, ...), Wikimedia pageviews, Google Trends, synthetic data
TiRex	Paper, Tables 5–6: subsets of the Chronos and GIFT-Eval training sets

None of them lists Australian retail turnover or US employment statistics. "Not listed" is not "not seen": a corpus can contain a series under another name. Say what you can and cannot rule out.

Where this goes wrong

- **Choosing on the validation window alone.** Four windows, and the scoreboard's mean — not window 1.
- **Reading one seed as the network's score.** A small network on one short series can move a lot from seed to seed. Run your best one twice.
- **"Foundation" read as "better".** On many Track B series ETS or ARIMA is still the one to beat.
- **Building the mix after reading the table.** Then it is a selection in disguise. Write 2.3's note first

- **Building the mix after reading the table.** Then it is a selection in disguise. Write 2.3 s note first.
- **Judging an interval on the windows it was built from.** `interval_check` leaves each window out.
- **Forecasting from `train`.** The submission calls every forecaster again on the whole of `y`.
- **An API key in the notebook.** TimeGPT is optional; if you use it, the key comes from a Colab secret.

📖 FPP3 §12.4 Neural network models · §13.4 Forecast combinations · Chronos-2 technical report · TimesFM 2.5 model card · TiRex paper

🖌 Section 4 — Code quality and reproducibility (1 point)

- The notebook runs top to bottom on a fresh kernel without errors.
- The submission file is written **by the notebook**, not edited by hand.
- No leakage: everything is compared on the rolling windows; only the final forecast sees all of `y`; no network is trained on any series but yours.
- Every number in your text comes from a cell.

⭐ Optional: TimeGPT

If you created a TimeGPT key for the final project (the Week 10 setup handout), you may add TimeGPT to the scoreboard as one more forecaster — the key from a Colab secret, never in the notebook. It counts toward the bonus for a justified step beyond the template, not toward the 8 points.

📤 Submission

Upload three files to the Week 10 activity on Moodle **before 23:59 today**:

- `SURNAME_lab10_forecast.csv` — written by the last cell of the template. Exactly h rows and these columns:

Column	What
`student_id`	your identifier
`code`	your series code, the same as before
`ds`	the dates you are forecasting, in order
`yhat`	your point forecast
`yhat_lo_80`, `yhat_hi_80`	your 80% interval

- `SURNAME_lab10.pdf` — the report, with the justification block;
- `SURNAME_lab10.ipynb` — the source.

⚠ **A file that cannot be read scores 0 for accuracy**

Exactly h rows, the dates asked for, no `NaN`, the interval around the point forecast. The template's check cell is the same one the scorer uses — if it passes, your file is fine.

⏱ How long this should take

Ninety minutes. The networks train 12 times (three models, four windows) — a few minutes on a CPU; the foundation models need seconds once downloaded. Most of the time should go into Section 3.